

Word Meaning and Similarity

Terminology: lemma and wordform

- A **lemma** or **citation form**
 - Same stem, part of speech, rough semantics
- A **wordform**
 - The inflected word as it appears in text

Wordform	Lemma
banks	bank
sung	sing

Lemmas have senses

- One lemma “bank” can have many meanings:

Sense 1: • ...a **bank**₁ can hold the investments in a custodial account...

Sense 2: • “...as agriculture burgeons on the east **bank**₂ of the river will shrink even more”

- **Sense (or word sense)**
 - A discrete representation of an aspect of a word’s meaning.
- The lemma **bank** here has two senses

Homonymy

Homonyms: words that share a form but have unrelated, distinct meanings:

- **bank₁**: financial institution, **bank₂**: sloping land
- **bat₁**: club for hitting a ball, **bat₂**: nocturnal flying mammal

Homonymy is of two types:

Homephones and Homographs

- Homophones: senses that are linked to lemmas with the same pronunciation but different spellings (e.g., wood/would, two/too/to, right/write)
- Homographs: distinct senses linked to lemmas with the same orthographic form but different pronunciation
 - *bass* - a deep voice or tone/a kind of fish
 - *bat* - a piece of sporting equipment used in baseball/a winged animal associated with vampires

Synonyms

- Word that have the same meaning in some or all contexts.
 - filbert / hazelnut
 - couch / sofa
 - big / large
 - automobile / car
 - vomit / throw up
 - Water / H₂O
- Two lexemes are synonyms
 - if they can be substituted for each other in all situations
 - If so they have the same **propositional meaning**

Synonymy is a relation between senses rather than words

- Consider the words *big* and *large*
- Are they synonyms?
 - How **big** is that plane?
 - Would I be flying on a **large** or small plane?
- How about here:
 - Miss Nelson became a kind of **big** sister to Benjamin.
 - ?Miss Nelson became a kind of **large** sister to Benjamin.
- Why?
 - *big* has a sense that means being older, or grown up
 - *large* lacks this sense

Antonyms

- Senses that are opposites with respect to one feature of meaning
- Otherwise, they are very similar!

dark/light short/long fast/slow rise/fall
hot/cold up/down in/out

- More formally: antonyms can
 - define a binary opposition
or be at opposite ends of a scale
 - long/short, fast/slow
 - Be **reversives**:
 - rise/fall, up/down

Hyponymy and Hypernymy

- One sense is a **hyponym** of another if the first sense is more specific, denoting a subclass of the other
 - *car* is a hyponym of *vehicle*
 - *mango* is a hyponym of *fruit*
- Conversely **hypernym/superordinate** (“hyper is super”)
 - *vehicle* is a **hypernym** of *car*
 - *fruit* is a hypernym of *mango*

Superordinate/hyper	vehicle	fruit	furniture
Subordinate/hyponym	car	mango	chair

Meronymy

- The part-whole relation
 - A *leg* is part of a *chair*; a *wheel* is part of a *car*.
- *Wheel* is a **meronym** of *car*, and *car* is a **holonym** of *wheel*.

Computing with a Thesaurus

WordNet

WordNet 3.0

- A hierarchically organized lexical database
- On-line thesaurus + aspects of a dictionary
 - Some other languages available or under development
 - (Arabic, Finnish, German, Portuguese...)

Category	Unique Strings
Noun	117,798
Verb	11,529
Adjective	22,479
Adverb	4,481

Senses of “bass” in Wordnet

Noun

- **S: (n) bass** (the lowest part of the musical range)
- **S: (n) bass**, **bass part** (the lowest part in polyphonic music)
- **S: (n) bass**, **basso** (an adult male singer with the lowest voice)
- **S: (n) sea bass**, **bass** (the lean flesh of a saltwater fish of the family Serranidae)
- **S: (n) freshwater bass**, **bass** (any of various North American freshwater fish with lean flesh (especially of the genus Micropterus))
- **S: (n) bass**, **bass voice**, **basso** (the lowest adult male singing voice)
- **S: (n) bass** (the member with the lowest range of a family of musical instruments)
- **S: (n) bass** (nontechnical name for any of numerous edible marine and freshwater spiny-finned fishes)

Adjective

- **S: (adj) bass**, **deep** (having or denoting a low vocal or instrumental range) *"a deep voice"; "a bass voice is lower than a baritone voice"; "a bass clarinet"*

How is “sense” defined in WordNet?

- The **synset (synonym set)**, the set of near-synonyms, instantiates a sense or concept, with a **gloss**
- Example: **chump** as a noun with the **gloss**:
“a person who is gullible and easy to take advantage of”
- This sense of “chump” is shared by 9 words:
chump¹, fool², gull¹, mark⁹, patsy¹, fall guy¹,
sucker¹, soft touch¹, mug²
- Each of **these** senses have this same gloss
 - (Not **every** sense; sense 2 of gull is the aquatic bird)

WordNet Hypernym Hierarchy for “bass”

- S: (n) bass, basso (an adult male singer with the lowest voice)
 - direct hypernym / inherited hypernym / sister term
 - S: (n) singer, vocalist, vocalizer, vocaliser (a person who sings)
 - S: (n) musician, instrumentalist, player (someone who plays a musical instrument (as a profession))
 - S: (n) performer, performing artist (an entertainer who performs a dramatic or musical work for an audience)
 - S: (n) entertainer (a person who tries to please or amuse)
 - S: (n) person, individual, someone, somebody, mortal, soul (a human being) *"there was too much for one person to do"*
 - S: (n) organism, being (a living thing that has (or can develop) the ability to act or function independently)
 - S: (n) living thing, animate thing (a living (or once living) entity)
 - S: (n) whole, unit (an assemblage of parts that is regarded as a single entity) *"how big is that part compared to the whole?"*; *"the team is a unit"*
 - S: (n) object, physical object (a tangible and visible entity; an entity that can cast a shadow) *"it was full of rackets, balls and other objects"*
 - S: (n) physical entity (an entity that has physical existence)
 - S: (n) entity (that which is perceived or known or inferred to have its own distinct existence (living or nonliving))

Distributional Similarity

Problems with thesaurus-based meaning

- We don't have a thesaurus for every language
- Even if we do, they have problems with **recall**
 - Many words are missing
 - Most (if not all) phrases are missing
 - Some connections between senses are missing
 - Thesauri work less well for verbs, adjectives
 - Adjectives and verbs have less structured hyponymy relations

Distributional models of meaning

- Also called vector-space models of meaning
- Offer much higher recall than hand-built thesauri
 - Although they tend to have lower precision
- Zellig Harris (1954): “**oculist** and **eye-doctor** ... occur in almost the same environments....
If A and B have almost identical environments we say that they are synonyms.
- Firth (1957): “You shall know a word by the company it keeps!”

Intuition of distributional word similarity

- Nida example:

A bottle of **tesgüino** is on the table
Everybody likes **tesgüino**
Tesgüino makes you drunk
We make **tesgüino** out of corn.

- From context words humans can guess **tesgüino** means
 - an alcoholic beverage like **beer**
- Intuition for algorithm:
 - Two words are similar if they have similar word contexts.

The Term-Context matrix

- Instead of using entire documents, use smaller contexts
 - Paragraph
 - Window of 10 words
- A word is now defined by a vector over counts of context words

Sample contexts: 20 words (Brown corpus)

- equal amount of sugar, a sliced lemon, a tablespoonful of **apricot** preserve or jam, a pinch each of clove and nutmeg,
 - on board for their enjoyment. Cautiously she sampled her first **pineapple** and another fruit whose taste she likened to that of
 - of a recursive type well suited to programming on the **digital** computer. In finding the optimal R-stage policy from that of
 - substantially affect commerce, for the purpose of gathering data and **information** necessary for the
- 21 study authorized in the first section of this

Term-context matrix for word similarity

- Two **words** are similar in meaning if their context vectors are similar

	aardvark	computer	data	pinch	result	sugar	...
apricot	0	0	0	1	0	1	
pineapple	0	0	0	1	0	1	
digital	0	2	1	0	1	0	
information	0	1	6	0	4	0	

Should we use raw counts?

- Raw word frequency is not a great measure of association between words
 - It's very skewed
 - “the” and “of” are very frequent, but maybe not the most discriminative
- We'd rather have a measure that asks whether a context word is **particularly informative** about the target word.

Should we use raw counts?

- For the term-document matrix
 - We used **tf-idf** instead of raw term counts
- For the term-context matrix
 - **Positive Pointwise Mutual Information (PPMI)** is common

Term frequency (tf)

$$tf_{t,d} = \text{count}(t,d)$$

Instead of using raw count, we squash a bit:

$$tf_{t,d} = \log_{10}(\text{count}(t,d)+1)$$

Document frequency (df)

df_t is the number of documents t occurs in.

(note this is not collection frequency: total count across all documents)

"*Romeo*" is very distinctive for one Shakespeare play:

	Collection Frequency	Document Frequency
Romeo	113	1
action	113	31

Inverse document frequency (idf)

$$\text{idf}_t = \log_{10} \left(\frac{N}{\text{df}_t} \right)$$

N is the total number of documents
in the collection

Word	df	idf
Romeo	1	1.57
salad	2	1.27
Falstaff	4	0.967
forest	12	0.489
battle	21	0.246
wit	34	0.037
fool	36	0.012
good	37	0
sweet	37	0

Final tf-idf weighted value for a word

$$w_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

Raw counts:

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	13
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

tf-idf:

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	0.074	0	0.22	0.28
good	0	0	0	0
fool	0.019	0.021	0.0036	0.0083
wit	0.049	0.044	0.018	0.022

Pointwise Mutual Information

- **Pointwise mutual information:**

- Do events x and y co-occur more than if they were independent?

$$\text{PMI}(X, Y) = \log_2 \frac{P(x, y)}{P(x)P(y)}$$

- **PMI between two words:** (Church & Hanks 1989)

- Do words x and y co-occur more than if they were independent?

$$\text{PMI}(\text{word}_1, \text{word}_2) = \log_2 \frac{P(\text{word}_1, \text{word}_2)}{P(\text{word}_1)P(\text{word}_2)}$$

- **Positive PMI between two words** (Niwa & Nitta 1994)

- Replace all PMI values less than 0 with zero

$$p_{ij} = \frac{f_{ij}}{\sum_{i=1}^W \sum_{j=1}^C f_{ij}}$$

apricot
pineapple
digital
information

Count(w,context)					
computer	data	pinch	result	sugar	
0	0	1	0	1	
0	0	1	0	1	
2	1	0	1	0	
1	6	0	4	0	

$$p(w=\text{information}, c=\text{data}) = 6/19 = .32$$

$$p(w=\text{information}) = 11/19 = .58$$

$$p(c=\text{data}) = 7/19 = .37$$

$$p(w_i) = \frac{\sum_{j=1}^C f_{ij}}{N}$$

$$p(c_j) = \frac{\sum_{i=1}^W f_{ij}}{N}$$

PPMI(w,context)

	computer	data	pinch	result	sugar
apricot	-	-	2.25	-	2.25
pineapple	-	-	2.25	-	2.25
digital	1.66	0.00	-	0.00	-
information	0.00	0.57	-	0.47	-

		p(w,context)					p(w)
		computer	data	pinch	result	sugar	
$pmi_{ij} = \log_2 \frac{p_{ij}}{p_{i*}p_{*j}}$	apricot	0.00	0.00	0.05	0.00	0.05	0.11
	pineapple	0.00	0.00	0.05	0.00	0.05	0.11
	digital	0.11	0.05	0.00	0.05	0.00	0.21
	information	0.05	0.32	0.00	0.21	0.00	0.58
	p(context)	0.16	0.37	0.11	0.26	0.11	

- $pmi(\text{information}, \text{data}) = \log_2 (.32 / (.37 * .58)) = .58$

(.57 using full precision)

	PPMI(w,context)				
	computer	data	pinch	result	sugar
apricot	-	-	2.25	-	2.25
pineapple	-	-	2.25	-	2.25
digital	1.66	0.00	-	0.00	-
information	0.00	0.57	-	0.47	-

Measuring similarity

- Given 2 target words v and w
- We'll need a way to measure their similarity.
- Most measure of vectors similarity are based on the:
- **Dot product** or **inner product** from linear algebra

$$\text{dot-product}(\vec{v}, \vec{w}) = \vec{v} \cdot \vec{w} = \sum_{i=1}^N v_i w_i = v_1 w_1 + v_2 w_2 + \dots + v_N w_N$$

- High when two vectors have large values in same dimensions.
- Low (in fact 0) for **orthogonal vectors** with zeros in complementary distribution

Problem with dot product

$$\text{dot-product}(\vec{v}, \vec{w}) = \vec{v} \cdot \vec{w} = \sum_{i=1}^N v_i w_i = v_1 w_1 + v_2 w_2 + \dots + v_N w_N$$

- Dot product is longer if the vector is longer. Vector length:

$$|\vec{v}| = \sqrt{\sum_{i=1}^N v_i^2}$$

- Vectors are longer if they have higher values in each dimension
- That means more frequent words will have higher dot products
- That's bad: we don't want a similarity metric to be sensitive to word frequency

Reminder: cosine for computing similarity

$$\cos(\vec{v}, \vec{w}) = \frac{\vec{v} \cdot \vec{w}}{|\vec{v}| |\vec{w}|} = \frac{\vec{v}}{|\vec{v}|} \cdot \frac{\vec{w}}{|\vec{w}|} = \frac{\sum_{i=1}^N v_i w_i}{\sqrt{\sum_{i=1}^N v_i^2} \sqrt{\sum_{i=1}^N w_i^2}}$$

v_i is the PPMI value for word v in context i
 w_i is the PPMI value for word w in context i .

$\vec{v} \rightarrow \vec{w}$
 $\text{Cos}(\vec{v}, \vec{w})$ is the cosine similarity of $\vec{v}$ and $\vec{w}$

Evaluating similarity (the same as for thesaurus-based)

- Intrinsic Evaluation:
 - Correlation between algorithm and human word similarity ratings
- Extrinsic (task-based, end-to-end) Evaluation:
 - Spelling error detection, WSD, essay grading
 - Taking TOEFL multiple-choice vocabulary tests

Levied is closest in meaning to which of these:
imposed, believed, requested, correlated

Word Representation



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Representing Words as Discrete Symbols

In traditional NLP, we regard words as discrete symbols:

hotel, conference, motel – a **localist** representation

Means one 1, the rest 0s

Such symbols for words can be represented by **one-hot** vectors:

motel = [0 0 0 0 0 0 0 0 0 0 1 0 0 0 0]

hotel = [0 0 0 0 0 0 0 1 0 0 0 0 0 0 0]

Vector dimension = number of words in vocabulary (e.g., 500,000+)



Problem with Words as Discrete Symbols

Example: in web search, if a user searches for “**Delhi motel**”, we would also like to match documents containing “**Delhi hotel**”

But:

motel = [0 0 0 0 0 0 0 0 0 0 1 0 0 0 0]

hotel = [0 0 0 0 0 0 0 1 0 0 0 0 0 0 0]

- These two vectors are orthogonal
- There is **no natural notion of similarity** for one-hot vectors!
- **Solution:**
 - Could try to rely on WordNet’s list of synonyms to get similarity?



Drawbacks of Thesaurus-based Approaches

- A useful resource but missing nuance
 - e.g., “proficient” is listed as a synonym for “good”: this is only correct in some contexts
 - Also, WordNet lists offensive synonyms in some synonym sets without any coverage of the connotations or appropriateness of words
- Missing new meanings of words
 - e.g., wicked, badass, nifty, wizard, genius, ninja, bombest
 - Impossible to keep up-to-date!
- Subjective
- Requires human labor to create and adapt



Prediction-based Methods



Word2vec

- Instead of **counting** how often each word w occurs near “*apricot*”
 - Train a classifier on a binary **prediction** task:
 - Is w likely to show up near “*apricot*”?
- We don’t actually care about this task
 - But we’ll take the learned classifier weights as the word embeddings
- Big idea: **Self-supervision**
 - A word c that occurs near *apricot* in the corpus acts as the gold “correct answer” for supervised learning
 - No need for human labels
 - Bengio et al. (2003); Collobert et al. (2011)

Adapted from NLP Lectures by Daniel Jurafsky



Approach: Predict if Candidate Word ' c ' is a "neighbor"

1. Treat the target word t and a neighboring context word c as **positive examples**.
2. Randomly sample other words in the lexicon to get negative examples
3. Use logistic regression to train a classifier to distinguish those two cases
4. Use the learned weights as the embeddings



Skip-Gram Training Data

Assume a **+/- 2 word window**, given training sentence:

... lemon, a [tablespoon of apricot jam, a] pinch ...

c_1

c_2

target

c_3

c_4

t



Skip-Gram Classifier

(assuming a +/- 2 word window)

... lemon, a [tablespoon of apricot jam, a] pinch ...

c_1 c_2 **target** c_3

- **Goal:** Train a classifier, that, **given a candidate (word, context) pair**

(apricot, jam)

(apricot, aardvark)

...

assigns each pair a probability:

$$P(+ | w, c)$$

$$P(- | w, c) = 1 - P(+ | w, c)$$



Similarity is Computed Using Dot Product

- **Remember:** *Two vectors are similar if they have a high dot product*
 - Cosine is just a normalized dot product
- **Similarity(w, c) $\propto w \cdot c$**
- We'll need to normalize to get a probability
 - Cosine isn't a probability either



Turning Dot Products into Probabilities

- $\text{Sim}(w, c) \approx w \cdot c$
- To turn this into a probability
 - We'll use the sigmoid function, as in **logistic regression**:

$$P(+|w, c) = \sigma(c \cdot w) = \frac{1}{1 + \exp(-c \cdot w)}$$

$$\begin{aligned} P(-|w, c) &= 1 - P(+|w, c) \\ &= \sigma(-c \cdot w) = \frac{1}{1 + \exp(c \cdot w)} \end{aligned}$$



How Skip-gram Classifier computes $P(+|w, c)$

$$P(+|w, c) = \sigma(c \cdot w) = \frac{1}{1 + \exp(-c \cdot w)}$$

- This is for one context word, but we have lots of context words.
- We'll assume independence and just multiply them:

$$P(+|w, c_{1:L}) = \prod_{i=1}^L \sigma(c_i \cdot w)$$

$$\log P(+|w, c_{1:L}) = \sum_{i=1}^L \log \sigma(c_i \cdot w)$$



Word2vec: Learning the Embeddings

Skip-Gram Training Data

Assume a **+/- 2 word window**, given training sentence:

... lemon, a [**tablespoon of** **apricot** jam, a] pinch ...

c_1

c_2

target

c_3

c_4

t

positive examples +

t	c
apricot	tablespoon
apricot	of
apricot	jam
apricot	a

For each positive example we'll
grab k negative examples, sampling
by frequency



Skip-Gram Training Data

Assume a **+/- 2 word window**, given training sentence:

... lemon, a [**tablespoon of** **apricot** jam, a] pinch ...

c_1

c_2

target

c_3

c_4

t

positive examples +

t	c
apricot	tablespoon
apricot	of
apricot	jam
apricot	a

negative examples -

t	c	t	c
apricot	aardvark	apricot	seven
apricot	my	apricot	forever
apricot	where	apricot	dear
apricot	coaxial	apricot	if



Loss Function for One w With $c_{pos}, c_{neg1} \dots c_{negk}$

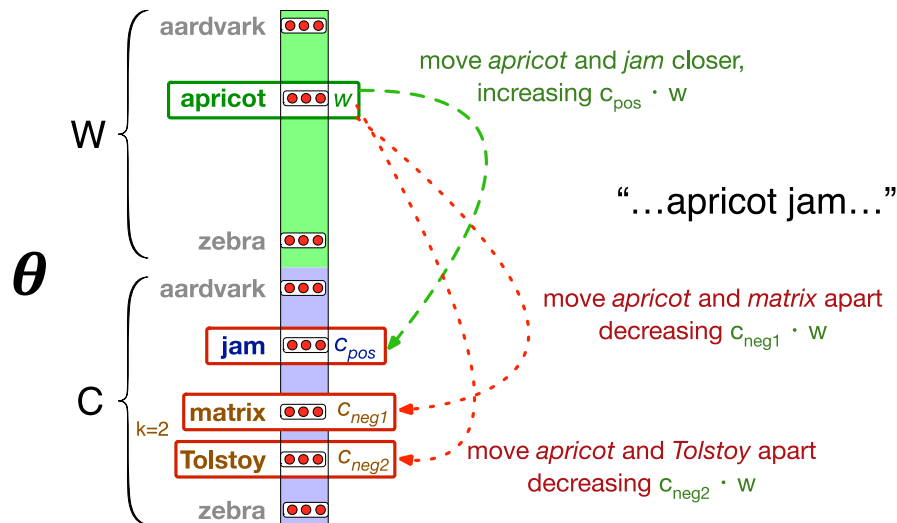
- Maximize the similarity of the target with the actual context words, and minimize the similarity of the target with the k negative sampled non-neighbor words.

$$\begin{aligned} L_{CE} &= -\log \left[P(+|w, c_{pos}) \prod_{i=1}^k P(-|w, c_{neg_i}) \right] \\ &= -\left[\log P(+|w, c_{pos}) + \sum_{i=1}^k \log P(-|w, c_{neg_i}) \right] \\ &= -\left[\log P(+|w, c_{pos}) + \sum_{i=1}^k \log (1 - P(+|w, c_{neg_i})) \right] \\ &= -\left[\log \sigma(c_{pos} \cdot w) + \sum_{i=1}^k \log \sigma(-c_{neg_i} \cdot w) \right] \end{aligned}$$



Learning the Classifier

- How to learn?
 - Stochastic gradient descent!
- We'll adjust the word weights to
 - make the positive pairs more likely
 - and the negative pairs less likely, over the entire training set.



The Derivatives of The Loss Function

$$L_{CE} = - \left[\log \sigma(c_{pos} \cdot w) + \sum_{i=1}^k \log \sigma(-c_{neg_i} \cdot w) \right]$$

$$\frac{\partial L_{CE}}{\partial c_{pos}} = [\sigma(c_{pos} \cdot w) - 1]w$$

$$\frac{\partial L_{CE}}{\partial c_{neg}} = [\sigma(c_{neg} \cdot w)]w$$

$$\frac{\partial L_{CE}}{\partial w} = [\sigma(c_{pos} \cdot w) - 1]c_{pos} + \sum_{i=1}^k [\sigma(c_{neg_i} \cdot w)]c_{neg_i}$$



Update Equation in SGD

Start with randomly initialized C and W matrices, then incrementally do updates

$$c_{pos}^{t+1} = c_{pos}^t - \eta [\sigma(c_{pos}^t \cdot w^t) - 1] w^t$$

$$c_{neg}^{t+1} = c_{neg}^t - \eta [\sigma(c_{neg}^t \cdot w^t)] w^t$$

$$w^{t+1} = w^t - \eta \left[[\sigma(c_{pos} \cdot w^t) - 1] c_{pos} + \sum_{i=1}^k [\sigma(c_{neg_i} \cdot w^t)] c_{neg_i} \right]$$



Some Interesting Results

Word Analogies

Test for linear relationships, examined by Mikolov et al. (2014)

a:b :: c:?

$$d = \arg \max_x \frac{(w_b - w_a + w_c)^T w_x}{\|w_b - w_a + w_c\|}$$

man:woman :: king:?

+ king [0.30 0.70]

- man [0.20 0.20]

+ woman [0.60 0.30]

queen [0.70 0.80]

